

SML Assignment 4

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Question 1: AdaBoost with Decision Stumps

Setup: Digits 4 and 9 were selected and mapped to -1 and $+1$. The data was flattened, normalized, and split into training and validation sets (1000 samples per class for validation).

PCA was applied using only the training data to reduce dimensionality to 5.

PCA Explanation: PCA projects the data onto directions of maximum variance by computing eigenvectors of the covariance matrix. Reducing to 5 dimensions helps remove noise and reduces computation while retaining most of the important structure.

Method: At each iteration, a decision stump is trained to minimize weighted classification error:

$$\epsilon_t = \sum w_i \mathbf{1}(y_i \neq h_t(x_i))$$

$$\alpha_t = \frac{1}{2} \log \frac{1 - \epsilon_t}{\epsilon_t}$$

$$w_i \leftarrow w_i \exp(-\alpha_t y_i h_t(x_i))$$

Final prediction:

$$H(x) = \text{sign} \left(\sum \alpha_t h_t(x) \right)$$

Each stump selects a feature and threshold that minimizes the weighted error. The weights ensure that misclassified samples receive more importance in subsequent iterations. This directly satisfies the requirement of adapting the model based on previous mistakes.

Result:

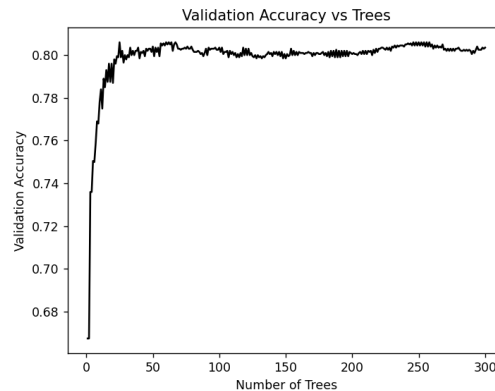


Figure 1: Validation Accuracy vs Number of Trees

Validation accuracy increases rapidly in early iterations and stabilizes later. This indicates that most useful patterns are captured early, and additional trees provide diminishing improvement.

Final Performance:

Best Iteration: 25

Train Accuracy: 0.805

Validation Accuracy: 0.806

Test Accuracy: 0.814

Question 2: Gradient Boosting

Method: Gradient Boosting minimizes squared loss:

$$L = \sum (y - \hat{y})^2$$

At each iteration, residuals are computed:

$$r_i = y_i - \hat{y}_i$$

A decision stump is fitted to these residuals and the model is updated:

$$\hat{y} \leftarrow \hat{y} + \eta h_t(x)$$

Instead of classification error, stumps are trained to minimize sum of squared residuals (SSR), which aligns with minimizing squared loss. Updating the labels using residuals corresponds to moving in the direction of the negative gradient.

Learning Rates:

$$\eta \in \{0.001, 0.01, 0.1, 0.2, 0.5, 1\}$$

Results:

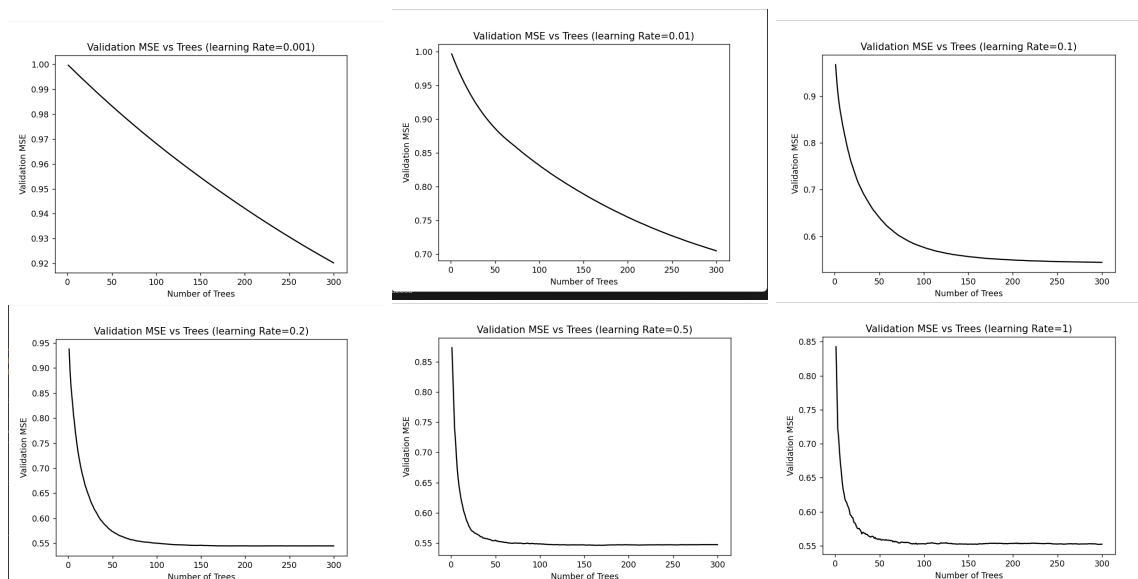


Figure 2: Validation MSE vs Trees for different learning rates

Smaller learning rates show slow but stable decrease in MSE. Larger learning rates reduce error quickly but often lead to overfitting.

As required, the variation across learning rates shows a trade-off between convergence speed and generalization. Lower η needs more trees but is more stable, while higher η risks overshooting and overfitting.

Observation: Gradient Boosting improves predictions by fitting residuals, making it directly tied to loss minimization. This is fundamentally different from AdaBoost, which reweights samples instead of optimizing a loss explicitly.

Question 3: Perceptron

Setup: Dataset A: $\mathcal{N}([-3, -3], I)$ and $\mathcal{N}([3, 3], I)$ Dataset B: Same means, covariance $3I$

Each dataset contains 200 samples per class and is split 70-30.

Method: Perceptron updates:

$$w \leftarrow w + \eta y_i x_i, \quad b \leftarrow b + \eta y_i$$

Prediction:

$$\hat{y} = \text{sign}(w^T x + b)$$

Training stops early if no misclassifications occur.

The algorithm tracks misclassification count per epoch and stops early when convergence is achieved. Decision boundaries are plotted to visualize separation between classes.

Results:

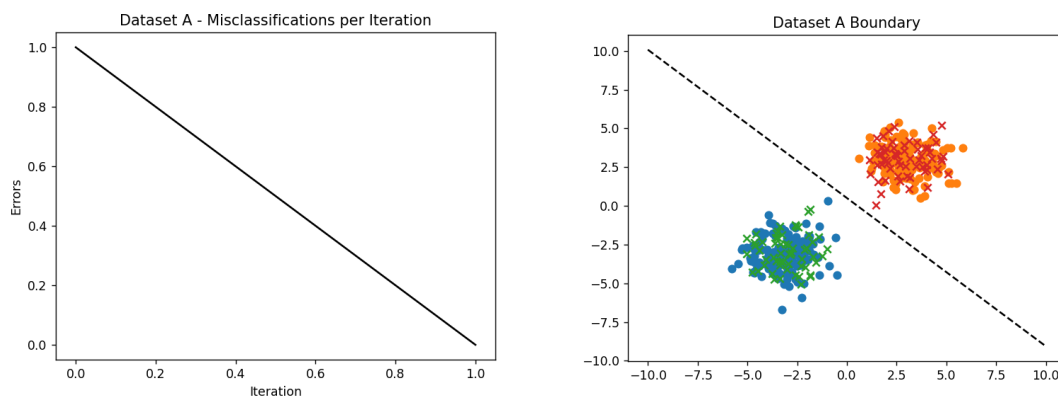


Figure 3: Dataset A

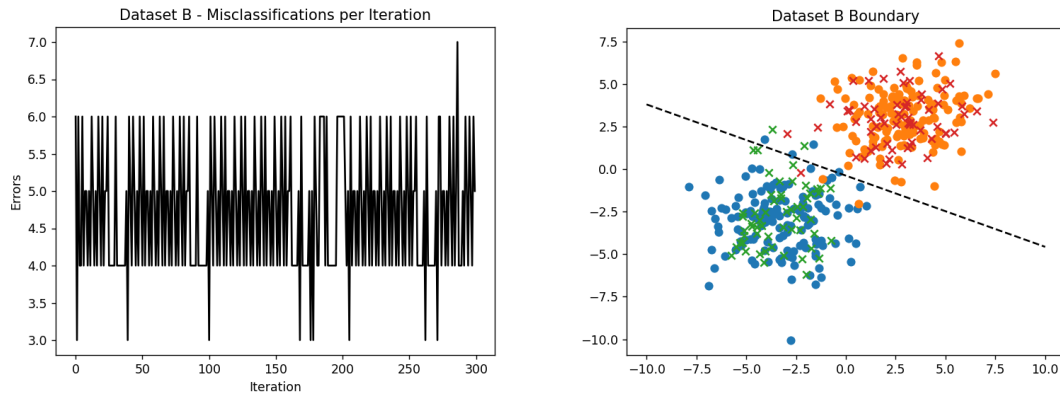


Figure 4: Dataset B

Dataset A converges quickly with zero misclassification, indicating linear separability. Dataset B does not converge due to overlap between classes.

Final Performance:

Dataset A Accuracy: 1.0

Dataset B Accuracy: 0.9917

As expected, the perceptron converges only for linearly separable data (Dataset A). For Dataset B, increased variance causes overlap, preventing convergence.

Observation: The results highlight that perceptron performance depends strongly on data distribution. Even small overlap can prevent convergence despite high accuracy.